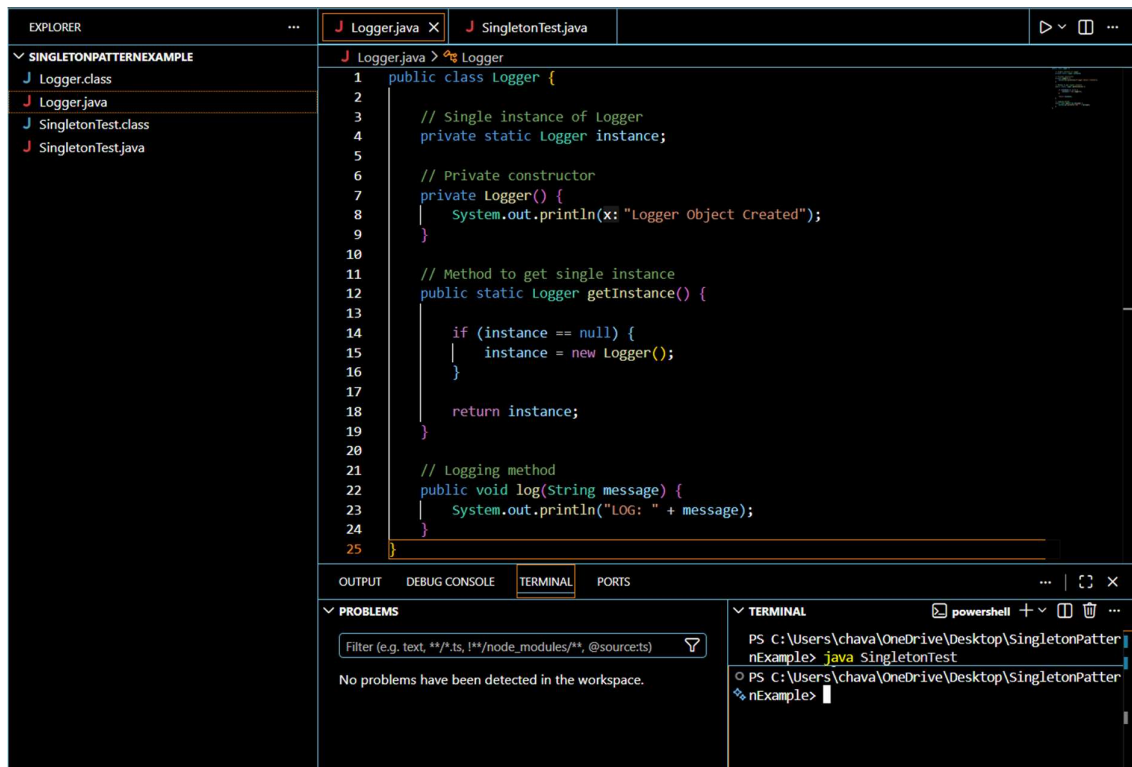


Exercise no. 1 Implementing the Singleton Pattern



```
1 public class Logger {
2
3     // Single instance of Logger
4     private static Logger instance;
5
6     // Private constructor
7     private Logger() {
8         System.out.println("Logger Object Created");
9     }
10
11     // Method to get single instance
12     public static Logger getInstance() {
13
14         if (instance == null) {
15             instance = new Logger();
16         }
17
18         return instance;
19     }
20
21     // Logging method
22     public void log(String message) {
23         System.out.println("LOG: " + message);
24     }
25 }
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

PROBLEMS

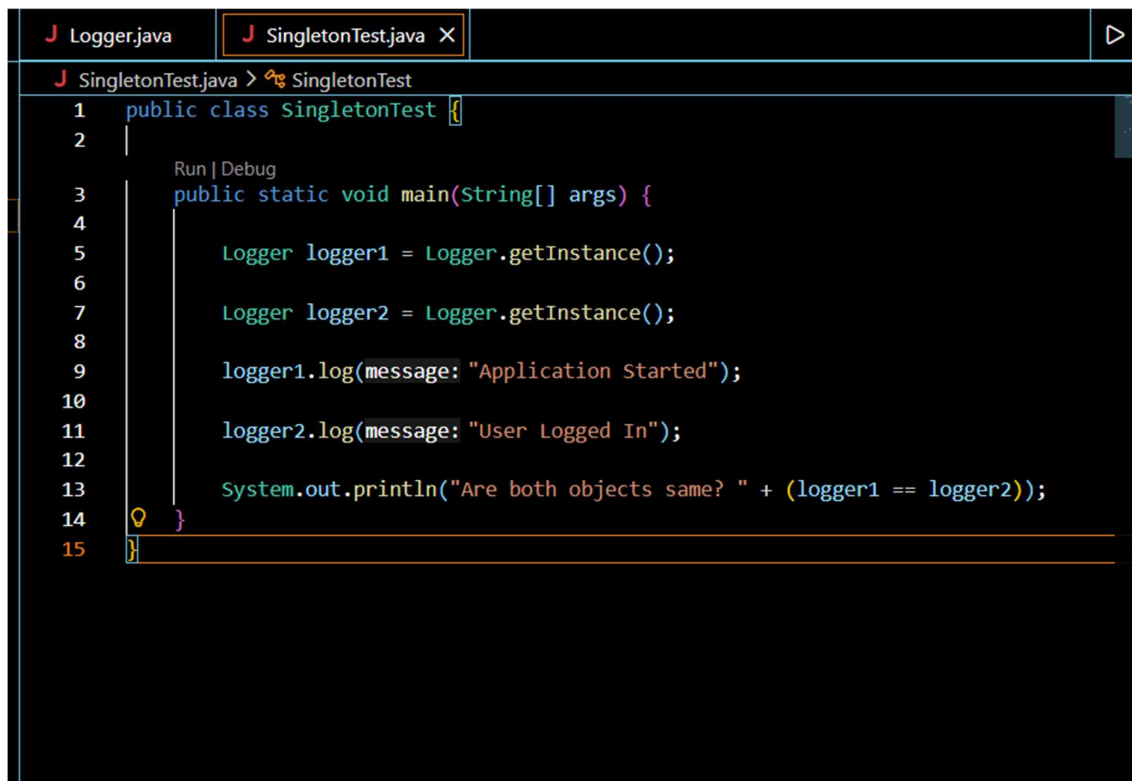
Filter (e.g. text, **/*.ts, !**/node_modules/**, @source:ts)

No problems have been detected in the workspace.

TERMINAL

PS C:\Users\chava\OneDrive\Desktop\SingletonPatternExample> java SingletonTest

PS C:\Users\chava\OneDrive\Desktop\SingletonPatternExample>



```
1 public class SingletonTest {
2
3     Run | Debug
4     public static void main(String[] args) {
5
6         Logger logger1 = Logger.getInstance();
7
8         Logger logger2 = Logger.getInstance();
9
10        logger1.log(message: "Application Started");
11
12        logger2.log(message: "User Logged In");
13
14        System.out.println("Are both objects same? " + (logger1 == logger2));
15    }
16 }
```

```
▼ TERMINAL powershell + v [ ] [ ] ...

PS C:\Users\chava\OneDrive\Desktop\SingletonPatternExample> javac Logger.j
ava SingletonTest.java
PS C:\Users\chava\OneDrive\Desktop\SingletonPatternExample> java Singleton
Test
Logger Object Created
LOG: Application Started
LOG: User Logged In
Are both objects same? true
PS C:\Users\chava\OneDrive\Desktop\SingletonPatternExample>
```

Exercise 2: Implementing the Factory Method Pattern

```
▼ FACTORYMETHODPATTERNEXA...
J Document.class
J Document.java
J DocumentFactory.class
J DocumentFactory.java
J ExcelDocument.class
J ExcelDocument.java
J ExcelFactory.class
J ExcelFactory.java
J FactoryTest.class
J FactoryTest.java
J PdfDocument.class
J PdfDocument.java
J PdfFactory.class
J PdfFactory.java
J WordDocument.class
J WordDocument.java
J WordFactory.class
J WordFactory.java
```

```
FactoryMethodPatternExample
J WordFactory.java J PdfFactory.java J ExcelFactory.java J FactoryTest.java
J FactoryTest.java > FactoryTest
1 public class FactoryTest {
2
3   Run | Debug
4   public static void main(String[] args) {
5
6     // Create Word document
7     DocumentFactory wordFactory = new WordFactory();
8
9     Document wordDoc = wordFactory.createDocument();
10
11    wordDoc.open();
12
13    // Create PDF document
14
15    DocumentFactory pdfFactory = new PdfFactory();
16
17    Document pdfDoc = pdfFactory.createDocument();
18
19    pdfDoc.open();
20
21    // Create Excel document
22
23    DocumentFactory excelFactory = new ExcelFactory();
24
25    Document excelDoc = excelFactory.createDocument();
26
27    excelDoc.open();
28
29  }
30
31 }
```

```
java J PdfDocument.java J ExcelDocument.java X J DocumentFactory
J ExcelDocument.java > ExcelDocument
1 public class ExcelDocument implements Document {
2
3   public void open() {
4
5     System.out.println("Opening Excel Document");
6
7   }
8
9 }
```

```
J DocumentFactory.java > DocumentFactory
1 public abstract class DocumentFactory {
2
3   public abstract Document createDocument();
4
5 }
```

```

J WordFactory.java > WordFactory
1 public class WordFactory extends DocumentFactory {
2
3 public Document createDocument() {
4
5 return new WordDocument();
6
7 }
8
9

```

```

J PdfFactory.java > PdfFactory
1 public class PdfFactory extends DocumentFactory {
2
3 public Document createDocument() {
4
5 return new PdfDocument();
6
7 }
8
9

```

```

J ExcelFactory.java > ExcelFactory
1 public class ExcelFactory extends DocumentFactory {
2
3 public Document createDocument() {
4
5 return new ExcelDocument();
6
7 }
8
9

```

```

J FactoryTest.java > FactoryTest
1 public class FactoryTest {
2
3     Run|Debug
4     public static void main(String[] args) {
5
6         // Create Word document
7         DocumentFactory wordFactory = new WordFactory();
8         Document wordDoc = wordFactory.createDocument();
9         wordDoc.open();
10
11        // Create PDF document
12        DocumentFactory pdfFactory = new PdfFactory();
13        Document pdfDoc = pdfFactory.createDocument();
14        pdfDoc.open();
15
16        // Create Excel document
17        DocumentFactory excelFactory = new ExcelFactory();
18        Document excelDoc = excelFactory.createDocument();
19        excelDoc.open();
20
21    }
22
23
24
25
26
27
28
29
30
31

```

```

v TERMINAL powershell + v
PS C:\Users\chava\OneDrive\Desktop\FactoryMethodPatternExample> javac *.java
PS C:\Users\chava\OneDrive\Desktop\FactoryMethodPatternExample> java FactoryTest
Opening Word Document
Opening PDF Document
Opening Excel Document
PS C:\Users\chava\OneDrive\Desktop\FactoryMethodPatternExample>

```